

Physical $X + \psi$ denominator from the Human-Note plumbing sum

Machine Note

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Status: theta- and glasses-channel resummations derived and implemented. The noncompact loop-momentum Gaussian is left as a direct plumbing Gaussian and is **not** replaced here by a canonical-frame spectral determinant. This note concerns the physical free boson and physical Majorana in the denominator of Q_L , not the auxiliary Majorana used in the double-Virasoro quotient. The Human Note is not modified.

1 Physical basis and the literal nonchiral sum

Use the theta edge order

$$e = (0, 1, \infty).$$

For one chiral physical free superfield, use the ordered Fock basis

$$|p; \lambda, S\rangle = a_{-\lambda_1} \cdots a_{-\lambda_k} \psi_{-r_1} \cdots \psi_{-r_s} |p\rangle, \quad r_1 > \cdots > r_s > 0,$$

where λ is an integer partition and $S \subset \mathbb{Z}_{>0} - \frac{1}{2}$. Define

$$N(\lambda, S) = |\lambda| + \sum_{r \in S} r, \quad \pi(\lambda, S) = |S| \pmod{2}, \quad h(p) = \frac{p^2}{2}.$$

Write $U_e = (\lambda_e, S_e)$, $V_e = (\mu_e, T_e)$ for the two ends selected by the inverse Gram matrix, with tilded letters for the antiholomorphic copy. After dividing by the connected target-space volume, the physical momentum measure is

$$[d\mathbf{p}]_\Theta = \prod_{e=0,1,\infty} dp_e \delta(p_0 + p_1 + p_\infty),$$

up to the fixed convention-dependent powers of 2π . The literal Human-Note theta prescription becomes

$$\begin{aligned} Z_{X+\psi}^{\Theta, \text{pl}} = \int [d\mathbf{p}]_\Theta \sum_{\mathbf{U}, \mathbf{V}, \tilde{\mathbf{U}}, \tilde{\mathbf{V}}} \prod_e q_e^{h(p_e) + N(U_e)} \bar{q}_e^{h(p_e) + N(\tilde{U}_e)} \eta_e^{\pi(U_e)} \bar{\eta}_e^{\pi(\tilde{U}_e)} \\ \times (-1)^{Q(\pi(\mathbf{U}) + \pi(\tilde{\mathbf{U}}))} \prod_e (\mathbb{B}_e^{-1})^{U_e \tilde{U}_e; V_e \tilde{V}_e} \\ \times \mathcal{R}_\Theta(\mathbf{U}, \tilde{\mathbf{U}}; \mathbf{p}) \mathcal{R}_\Theta(\mathbf{V}, \tilde{\mathbf{V}}; \mathbf{p}), \end{aligned} \quad (1.1)$$

where

$$Q(x_0, x_1, x_\infty) = x_0 x_1 + x_0 x_\infty + x_1 x_\infty \pmod{2}. \quad (1.2)$$

Here $\mathbb{B}_e^{-1}$ is the inverse Gram matrix of the **full nonchiral physical** $X + \psi$ state. Equation (1.1), rather than a theta constant or a spectral determinant, is the definition of the denominator in the theta plumbing Weyl frame.

The sign in (1.1) is not yet a product of a holomorphic and an antiholomorphic sign. The inverse tensor-product Gram matrix and the two full pants correlators must be factorized before making such a statement.

2 What the analytic free-field three-point function implies

Normalize

$$X(Z_1)X(Z_2) \sim -\log Z_{12}, \quad Z_{ij} = z_i - z_j - \theta_i \theta_j,$$

and let

$$\mathcal{V}_p(Z) =: e^{ipX(Z)} :, \quad h(p) = \frac{p^2}{2}.$$

The holomorphic primary three-point function is

$$\langle \mathcal{V}_{p_\infty}(Z_\infty) \mathcal{V}_{p_1}(Z_1) \mathcal{V}_{p_0}(Z_0) \rangle = \delta(p_0 + p_1 + p_\infty) \prod_{i < j} Z_{ij}^{p_i p_j}. \quad (2.1)$$

It contains the even superconformal three-point structure and no term proportional to the odd invariant Θ_{123} . Therefore

$$C_{p_0 p_1 p_\infty}^{(1)} = 0. \quad (2.2)$$

This fact is what makes the free-theory reduction simpler than the super-Liouville numerator. It must be used only after the full Human-Note sign algebra. Here is that factorization explicitly. Set

$$r_e = \pi(U_e) = \pi(V_e), \quad \tilde{r}_e = \pi(\tilde{U}_e) = \pi(\tilde{V}_e).$$

The equalities follow because the inverse Gram matrix connects states at the same level and hence with the same fermion parity. The physical exponential primaries are even. Specializing the Human-Note tensor-product BPZ rule to these free primaries gives

$$(\mathbb{B}_e^{-1})^{U_e \tilde{U}_e; V_e \tilde{V}_e} = (-1)^{r_e \tilde{r}_e} (B_e^{-1})^{U_e V_e} (\tilde{B}_e^{-1})^{\tilde{U}_e \tilde{V}_e}. \quad (2.2a)$$

Each full pants correlator separately has a Koszul sign from moving all antiholomorphic operators to the right. The two pants correlators have the same edge parities, so these two signs square to one. With the normalized free primary coefficient stripped off, their product is therefore

$$\begin{aligned} \mathcal{R}_\Theta(\mathbf{U}, \tilde{\mathbf{U}}; \mathbf{p}) \mathcal{R}_\Theta(\mathbf{V}, \tilde{\mathbf{V}}; \mathbf{p}) \\ = \rho(\mathbf{U}; \mathbf{p}) \rho(\mathbf{V}; \mathbf{p}) \tilde{\rho}(\tilde{\mathbf{U}}; \mathbf{p}) \tilde{\rho}(\tilde{\mathbf{V}}; \mathbf{p}). \end{aligned} \quad (2.2b)$$

Consequently the exponent of the complete sign in (1.1) is

$$\epsilon_{\text{tot}} = Q(\mathbf{r} + \tilde{\mathbf{r}}) + \mathbf{r} \cdot \tilde{\mathbf{r}}. \quad (2.2c)$$

For three edges the elementary identity

$$Q(\mathbf{r} + \tilde{\mathbf{r}}) + \mathbf{r} \cdot \tilde{\mathbf{r}} = Q(\mathbf{r}) + Q(\tilde{\mathbf{r}}) + (r_0 + r_1 + r_\infty)(\tilde{r}_0 + \tilde{r}_1 + \tilde{r}_\infty) \pmod{2} \quad (2.2d)$$

does all the remaining work. The holomorphic and antiholomorphic pants selection rules require

$$a = r_0 + r_1 + r_\infty = \tilde{r}_0 + \tilde{r}_1 + \tilde{r}_\infty \pmod{2}. \quad (2.2e)$$

The last term of (2.2d) is therefore $a^2 = a$. Thus the full nonchiral sign becomes

$$(-1)^{a+Q(\mathbf{r})+Q(\tilde{\mathbf{r}})}. \quad (2.3)$$

Because (2.2) forces $a = 0$, the complete nonchiral sum is

$$Z_{X+\psi}^{\Theta, \text{pl}} = \int [d\mathbf{p}]_\Theta \left| \mathcal{F}_{X+\psi, 0}^{\Theta, \text{pl}}(\mathbf{p}; \mathbf{q}, \boldsymbol{\eta}) \right|^2. \quad (2.4)$$

Thus holomorphic factorization is a result of (2.1)–(2.3), not an assumption made in (1.1). In particular, the descendant sign inside the chiral block remains

$$(-1)^{Q(\pi_0, \pi_1, \pi_\infty)}. \quad (2.5)$$

3 Analytic physical-Majorana pants tensor

Let the positive integer m label the NS mode

$$\psi_{-(m-1/2)}|0\rangle.$$

Use slot order $(\infty, 1, 0)$, and write a combined index as $\alpha = (s, m)$. The analytic sphere contraction matrix is the antisymmetric matrix K whose independent entries are

$$\begin{aligned} K_{(\infty, m), (1, n)} &= \begin{cases} \binom{m-1}{n-1}, & m \geq n, \\ 0, & m < n, \end{cases} \\ K_{(\infty, m), (0, n)} &= \delta_{mn}, \\ K_{(1, m), (0, n)} &= (-1)^{m-1} \binom{m+n-2}{m-1}. \end{aligned} \tag{3.1}$$

For three Fock states S_∞, S_1, S_0 , let $I = S_\infty \sqcup S_1 \sqcup S_0$ be the corresponding set of combined indices. In the generating-function order where modes increase on all three slots, Wick's theorem gives

$$\hat{\rho}_\psi(S_\infty, S_1, S_0) = \text{Pf } K_I. \tag{3.2}$$

The physical BPZ bra reverses its fermion fields. Consequently the physical pants coefficient used in the literal state sum is

$$\rho_\psi(S_\infty, S_1, S_0) = (-1)^{|S_\infty|(|S_\infty|-1)/2} \hat{\rho}_\psi(S_\infty, S_1, S_0). \tag{3.3}$$

Both coefficients vanish when the total number of fermions is odd. In the theta graph the two identical pants coefficients are multiplied, so the BPZ reversal sign in (3.3) squares to one. In the glasses graph it is retained by the half-edge pairing ledger in Section 7.

4 Fredholm resummation of the Human-Note descendant sign

Introduce the plumbing half-variables

$$x_e = \eta_e \sqrt{q_e}. \tag{4.1}$$

For $\sigma = (\sigma_\infty, \sigma_1, \sigma_0) \in \{\pm 1\}^3$, define the diagonal mode-weight operator

$$(W_\sigma)_{(e, m), (e, m)} = \sigma_e x_e^{2m-1}, \tag{4.2}$$

and the Fredholm determinant

$$D_\sigma = \det_{\mathbb{F}}(1 + KW_\sigma). \tag{4.3}$$

The principal-minor identity and (3.2) give

$$D_\sigma = \sum_{S_\infty, S_1, S_0} \hat{\rho}_\psi(S_\infty, S_1, S_0)^2 \prod_e \sigma_e^{|S_e|} x_e^{2N(S_e)}. \tag{4.4}$$

Only parity patterns (000, 110, 101, 011) occur. On these four patterns, the Human-Note sign (2.5) is respectively $(+, -, -, -)$. Its exact all-level projection is consequently

$$\boxed{\mathcal{F}_\psi^{\Theta, \text{pl}} = \frac{1}{2} [-D_{(+,+,+)} + D_{(-,+,+)} + D_{(+,-,+)} + D_{(+,+, -)}]}. \quad (4.5)$$

This is the desired physical-fermion resummation. It uses the plumbing coordinates, the physical lifts, and the analytic pants tensor only. No period matrix occurs.

At a mode cutoff M , (4.5) requires four determinants of size $3M$. The cost is $O(M^3)$, instead of enumerating an exponentially growing collection of Fock states. Numerically one should use

$$\det(1 + KW) = \det(1 + W^{1/2}KW^{1/2}) \quad (4.6)$$

to balance the growing binomial entries of K against the decaying plumbing weights.

5 Charged boson and complete one-superfield resummation

The bosonic oscillator pants sum exponentiates by Wick's theorem. In the same theta plumbing marking its all-level resummation is

$$P_X^{\Theta, \text{pl}}(\mathbf{q}) = \prod_{[\gamma] \in \mathcal{P}_{\text{prim}}} \prod_{n=1}^{\infty} (1 - k_\gamma^n)^{-1}, \quad (5.1)$$

where every multiplier k_γ is computed from the Schottky generators constructed from the same $(0, 1, \infty)$ plumbing maps. Equation (5.1) is a resummation of the Heisenberg pants state sum; it is not a canonical Laplacian determinant.

The charged tensor is just as explicit. In the bra convention

$$\langle \alpha_\infty | V_{\alpha_1}(1) | \alpha_0 \rangle, \quad \alpha_\infty = \alpha_1 + \alpha_0,$$

divide every current mode by the square root of its mode number. In slot order $(\infty, 1, 0)$, the symmetric normalized current-current kernel K_X is obtained from

$$\begin{aligned} C_{\infty 1}(m, n) &= m \binom{m-1}{n-1} \mathbf{1}_{m \geq n}, \\ C_{\infty 0}(m, n) &= m \delta_{mn}, \\ C_{10}(m, n) &= (-1)^{m-1} n \binom{n+m-1}{m-1} \end{aligned} \quad (5.2)$$

by $(K_X)_{(e,m),(f,n)} = C_{ef}(m, n)/\sqrt{mn}$. The normalized current-exponential contractions form the linear vector

$$\ell_{\infty, m} = \frac{\alpha_1}{\sqrt{m}}, \quad \ell_{1, m} = \frac{(-1)^{m-1} \alpha_0}{\sqrt{m}}, \quad \ell_{0, m} = -\frac{\alpha_1}{\sqrt{m}}. \quad (5.3)$$

Thus the analytic charged pants generating function is

$$\exp\left(\frac{1}{2} t^T K_X t + \ell^T t\right). \quad (5.4)$$

This formula includes both pairwise current contractions and the singleton contractions of a current with one of the exponential primaries. Expanding (5.4) gives the charged Fock-module tensor $\rho_{\alpha_\infty, \alpha_1, \alpha_0}(\lambda, \mu, \nu)$.

At mode cutoff M , let

$$W_{(e,m),(e,m)} = q_e^m, \quad D = W^{1/2}, \quad B = DK_X D. \quad (5.5)$$

The inverse-BPZ sewing convention gives the same numerical charge labels to the two pants tensors. Contracting the two Gaussian generating functions therefore gives

$$P_{X,M}^{\Theta,\text{pl}} = \det(1 - B^2)^{-1/2}, \quad E_M(\alpha_0, \alpha_1) = (D\ell)^T(1 - B)^{-1}(D\ell). \quad (5.6)$$

The branch of the determinant is fixed by $P_{X,M} \rightarrow 1$ at the plumbing cusp. The complete fixed-charge chiral boson block is

$$\mathcal{F}_{X,M}^{\Theta,\text{pl}} = P_{X,M}^{\Theta,\text{pl}} \prod_{e=0,1,\infty} q_e^{\alpha_e^2/2} \exp E_M(\alpha_0, \alpha_1), \quad \alpha_\infty = -(\alpha_0 + \alpha_1) \quad (5.7)$$

when all three propagating charges are written as outgoing charges. The bra label in (5.3) is then $-\alpha_\infty = \alpha_0 + \alpha_1$.

Equations (5.5)–(5.7) are the requested direct resummation of the charged free-boson state sum. In particular, $P_{X,M}$ converges to (5.1) without using a period matrix.

The nonchiral charge dependence is a two-dimensional Gaussian. Define its quadratic form directly by

$$\left| \frac{\mathcal{F}_X^{\Theta,\text{pl}}(\alpha_0, \alpha_1; \mathbf{q})}{P_X^{\Theta,\text{pl}}(\mathbf{q})} \right|^2 = \exp \left[-\pi(\alpha_0, \alpha_1) A_\Theta^{\text{pl}}(\mathbf{q}) \begin{pmatrix} \alpha_0 \\ \alpha_1 \end{pmatrix} \right]. \quad (5.8)$$

The entries of A_Θ^{pl} are read directly from (5.7), or equivalently from its values at charge vectors $(1, 0), (0, 1), (1, 1)$. With completeness measure $d\alpha_0 d\alpha_1$,

$$\mathcal{G}_X^{\Theta,\text{pl}} = \int_{\mathbb{R}^2} d^2\alpha e^{-\pi\alpha^T A_\Theta^{\text{pl}} \alpha} = \det A_\Theta^{\text{pl}-1/2}. \quad (5.9)$$

No period matrix is used in (5.2)–(5.9). A rescaling of the momentum variable must rescale both A_Θ^{pl} and the completeness measure. This makes visible the otherwise hidden factor-of-two distinction between a vertex charge with $h = \alpha^2/2$ and a commonly used momentum variable $p = \sqrt{2}\alpha$ with $h = p^2/4$.

Combining (4.5), (5.1), and (5.9) gives the period-matrix-free plumbing formula for one physical free superfield:

$$\boxed{Z_{X+\psi}^{\Theta,\text{pl}} = \mathcal{G}_X^{\Theta,\text{pl}} \left| P_X^{\Theta,\text{pl}} \right|^2 \left| \mathcal{F}_\psi^{\Theta,\text{pl}} \right|^2.} \quad (5.10)$$

The $\widehat{c} = 9$ physical denominator is

$$Z_{\text{free}, \widehat{c}=9}^{\Theta,\text{pl}} = \left(Z_{X+\psi}^{\Theta,\text{pl}} \right)^9. \quad (5.11)$$

No auxiliary double-Virasoro fermion occurs in (1.1)–(5.11).

6 Relation to the theta-constant shortcut

If one separately proves, in precisely this plumbing trivialization and with the Human-Note lifts, that

$$\left(\mathcal{F}_\psi^{\Theta, \text{pl}}\right)^2 = \vartheta[\delta](0|\Omega_{\text{native}})P_X^{\Theta, \text{pl}}, \quad (6.1)$$

then (5.10) can be algebraically rewritten in the earlier theta-constant form. Equation (6.1) is not assumed in the construction above. It is now a cross-check between two resummations, so a possible determinant-line or marking mistake cannot silently define the physical denominator.

7 Glasses-channel resummation

For the glasses edge order (L, R, B) , direct chiral sewing gives

$$\begin{aligned} \mathcal{F}_\psi^{\text{Gl, pl}} = & \sum_{S_L, S_R, S_B} (-1)^{\pi_B(\pi_L + \pi_R)} \prod_{e=L, R, B} x_e^{2N(S_e)} \\ & \times \rho_\psi(S_L, S_B, S_L) \rho_\psi(S_R, S_B, S_R). \end{aligned} \quad (7.1)$$

Each nonzero pants tensor requires an even total number of fermions. Since the self-loop state occurs twice, this forces $\pi_B = 0$. Thus the explicit Human-Note glasses sign in (7.1) is one on the support of the free-Majorana tensor. It has not been discarded before this selection rule is applied.

Let

$$K_{\psi, \Gamma} = K \oplus K, \quad w_{e, m} = x_e^{2m-1}, \quad (7.2)$$

and put $\sqrt{w_{e, m}}$ on each of the two half-edges incident on edge e ; their diagonal matrix is D_ψ . Let J_ψ be the antisymmetric half-edge pairing matrix. In the half-edge order $(\infty_L, 1_L, 0_L, \infty_R, 1_R, 0_R)$, its nonzero edge orientations are

$$J_{\psi, L} = -1, \quad J_{\psi, R} = -1, \quad J_{\psi, B} = i. \quad (7.3)$$

The two minus signs are the self-loop bends. The bridge phase is the coefficient-basis correction: contracting two Grassmann generating functions gives $(-1)^{s(s-1)/2}$, while the surviving bridge occupations have even s . Therefore the factor $i^s = (-1)^{s/2}$ in (7.3) cancels it. The choice $-i$ gives the same result because odd bridge occupation vanishes. The exact cutoff resummation is

$$\boxed{\mathcal{F}_{\psi, M}^{\text{Gl, pl}} = \det(1 + J_\psi D_\psi K_{\psi, \Gamma} D_\psi)^{1/2}}. \quad (7.4)$$

The square-root branch is the one approaching one at the plumbing cusp. Equation (7.4) was checked against the literal state sum (7.1), not inferred from a theta constant.

For the boson, set $K_{X, \Gamma} = K_X \oplus K_X$, put $\sqrt{q_e^m}$ on each half-edge in D_X , and let J_X be the symmetric pairing with entry one across all three edges. Then

$$P_{X, M}^{\text{Gl, pl}} = \det(1 - J_X D_X K_{X, \Gamma} D_X)^{-1/2}. \quad (7.5)$$

For self-loop charges (α_L, α_R) , the concatenated source is $\ell_\Gamma = (\ell(\alpha_L, 0), \ell(\alpha_R, 0))$. With $j = D_X \ell_\Gamma$, the charged exponent and primary propagation are

$$E_M^{\text{Gl}} = \frac{1}{2} j^T (1 - J_X D_X K_{X, \Gamma} D_X)^{-1} J_X j, \quad q_L^{\alpha_L^2/2} q_R^{\alpha_R^2/2}. \quad (7.6)$$

The bridge primary is neutral. Reading the real quadratic form $A_{\text{Gl}}^{\text{pl}}$ from the absolute square of (7.6) gives

$$\mathcal{G}_X^{\text{Gl,pl}} = \det A_{\text{Gl}}^{\text{pl} - 1/2}, \quad (7.7)$$

in the displayed $d\alpha_L d\alpha_R$ measure. Hence

$$\boxed{Z_{X+\psi}^{\text{Gl,pl}} = \mathcal{G}_X^{\text{Gl,pl}} \left| P_X^{\text{Gl,pl}} \right|^2 \left| \mathcal{F}_\psi^{\text{Gl,pl}} \right|^2}. \quad (7.8)$$

Both (5.10) and (7.8) are raw plumbing-frame quantities. Their numerical values should not be equated without the channel-change/Weyl factors that belong to the full comparison.

8 Unpacking every factor in the physical denominator

The compact notation in (5.10) and (7.8) contains three logically different objects. For one real physical superfield they are

$$\underbrace{\mathcal{G}_X^{\text{pl}}}_{\text{two physical loop-charge integrals}} \times \underbrace{|P_X^{\text{pl}}|^2}_{\text{nonzero bosonic oscillators}} \times \underbrace{|\mathcal{F}_\psi^{\text{pl}}|^2}_{\text{physical NS-Majorana oscillators}}. \quad (8.1)$$

The last factor is the physical partner ψ of X . It is not the auxiliary Majorana introduced in the double-Virasoro branching identity.

8.1 Theta graph

At mode cutoff M , define

$$\Delta_{\psi,\Theta,M} = -D_{(+,+,+),M} + D_{(-,+,+),M} + D_{(+,-,+),M} + D_{(+,+, -),M}, \quad (8.2)$$

where each $D_{\sigma,M}$ is the $3M$ -dimensional determinant in (4.3), evaluated in its balanced form (4.6). Then

$$|\mathcal{F}_{\psi,M}^{\Theta,\text{pl}}|^2 = \frac{1}{4} |\Delta_{\psi,\Theta,M}|^2. \quad (8.3)$$

For the physical boson, let

$$B_M = D_M K_{X,M} D_M, \quad D_M^2 = \text{diag}_{(e,m)}(q_e^m). \quad (8.4)$$

The nonzero-mode factor is therefore

$$|P_{X,M}^{\Theta,\text{pl}}|^2 = |\det(1 - B_M^2)|^{-1}. \quad (8.5)$$

It remains to unpack the zero-mode Gaussian. Put $u = (\alpha_0, \alpha_1)^T$, and introduce the charge-incidence matrix

$$R_\Theta = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \end{pmatrix}, \quad R_\Theta u = (\alpha_0, \alpha_1, \alpha_\infty)^T. \quad (8.6)$$

Let $\Lambda_{\Theta,M}$ be the $3M$ -by-2 matrix defined by $\ell = \Lambda_{\Theta,M} u$. Its rows in slot order $(\infty, 1, 0)$ are

$$(\Lambda_{\Theta,M})_{(\infty,m)} = \frac{1}{\sqrt{m}}(0,1), \quad (\Lambda_{\Theta,M})_{(1,m)} = \frac{1}{\sqrt{m}}((-1)^{m-1},0), \quad (\Lambda_{\Theta,M})_{(0,m)} = \frac{1}{\sqrt{m}}(0,-1). \quad (8.7)$$

Writing $L_{\Theta,M} = D_M \Lambda_{\Theta,M}$, the charged oscillator exponent in (5.6) is

$$E_M(u) = u^T H_{\Theta,M} u, \quad H_{\Theta,M} = L_{\Theta,M}^T (1 - B_M)^{-1} L_{\Theta,M}. \quad (8.8)$$

Here and below only the symmetric part of H contributes to $u^T H u$. Taking the absolute square of the primary propagation and (8.8) gives the loop matrix explicitly:

$$A_{\Theta,M}^{\text{pl}} = -\frac{1}{\pi} [R_{\Theta}^T \text{diag}(\log |q_0|, \log |q_1|, \log |q_{\infty}|) R_{\Theta} + 2 \text{Re Sym } H_{\Theta,M}]. \quad (8.9)$$

Thus the charge integral is not an independent canonical-frame input:

$$\mathcal{G}_{X,M}^{\Theta,\text{pl}} = \det A_{\Theta,M}^{\text{pl}}^{-1/2}. \quad (8.10)$$

Substituting (8.3), (8.5), and (8.10) completely removes the shorthand from the one-superfield answer:

$$Z_{X+\psi,M}^{\Theta,\text{pl}} = \frac{|\Delta_{\psi,\Theta,M}|^2}{4 \sqrt{|\det A_{\Theta,M}^{\text{pl}}| |\det(1 - B_M^2)|}}. \quad (8.11)$$

For nine independent physical superfields,

$$Z_{\text{free}, \widehat{c}=9,M}^{\Theta,\text{pl}} = 2^{-18} (\det A_{\Theta,M}^{\text{pl}})^{-9/2} |\det(1 - B_M^2)|^{-9} |\Delta_{\psi,\Theta,M}|^{18}. \quad (8.12)$$

All determinants in (8.11)–(8.12) are constructed from the theta plumbing coordinates, lifts, and analytic pants kernels.

8.2 Glasses graph

Define the two cutoff matrices

$$\mathbb{M}_{\psi,\text{Gl},M} = 1 + J_{\psi} D_{\psi} K_{\psi,\Gamma,M} D_{\psi}, \quad \mathbb{M}_{X,\text{Gl},M} = 1 - J_X D_X K_{X,\Gamma,M} D_X. \quad (8.13)$$

Equations (7.4) and (7.5) then give

$$|\mathcal{F}_{\psi,M}^{\text{Gl},\text{pl}}|^2 = |\det \mathbb{M}_{\psi,\text{Gl},M}|, \quad |P_{X,M}^{\text{Gl},\text{pl}}|^2 = |\det \mathbb{M}_{X,\text{Gl},M}|^{-1}. \quad (8.14)$$

Put $u = (\alpha_L, \alpha_R)^T$. Let $\Lambda_{\text{Gl},M}$ be the $6M$ -by-2 matrix determined by

$$\ell_{\Gamma} = (\ell(\alpha_L, 0), \ell(\alpha_R, 0)) = \Lambda_{\text{Gl},M} u, \quad L_{\text{Gl},M} = D_X \Lambda_{\text{Gl},M}. \quad (8.15)$$

The matrix representing the exponent (7.6) is

$$H_{\text{Gl},M} = \frac{1}{2} L_{\text{Gl},M}^T \mathbb{M}_{X,\text{Gl},M}^{-1} J_X L_{\text{Gl},M}. \quad (8.16)$$

The bridge primary is neutral, so the glasses loop matrix is

$$A_{\text{Gl},M}^{\text{pl}} = -\frac{1}{\pi} [\text{diag}(\log |q_L|, \log |q_R|) + 2 \text{Re Sym } H_{\text{Gl},M}]. \quad (8.17)$$

Consequently the fully unpacked glasses expressions are

$$Z_{X+\psi,M}^{\text{Gl,pl}} = \frac{|\det \mathbb{M}_{\psi,\text{Gl},M}|}{\sqrt{\det A_{\text{Gl},M}^{\text{pl}} |\det \mathbb{M}_{X,\text{Gl},M}|}}, \quad (8.18)$$

and

$$Z_{\text{free},\hat{c}=9,M}^{\text{Gl,pl}} = (\det A_{\text{Gl},M}^{\text{pl}})^{-9/2} |\det \mathbb{M}_{\psi,\text{Gl},M}|^9 |\det \mathbb{M}_{X,\text{Gl},M}|^{-9}. \quad (8.19)$$

8.3 Normalization ledger

Equations (8.11)–(8.19) use the following precise normalization:

1. $XX \sim -\log z$ and $h(\alpha) = \alpha^2/2$.
2. The two independent genus-two charges are integrated with $d^2\alpha$, after dividing out the connected target-space volume.
3. The normalized free-field pants coefficient is one.
4. The plumbing propagator is q^{L_0} ; the common cylinder Casimir factor $q^{-c/24}$ is stripped.
5. No channel-change Weyl factor, determinant-line trivialization factor, or auxiliary double-*Virasoro* fermion is included.

If the completeness convention is $d\alpha/(2\pi)$ for every real charge, there are two independent charges for each of nine superfields, and the right-hand sides of (8.12) and (8.19) acquire the overall factor

$$(2\pi)^{-18}. \quad (8.20)$$

If instead the propagator convention includes the cylinder vacuum energy, one real $X + \psi$ superfield has $c = 3/2$. Its nonchiral multiplier is

$$\prod_{e=1}^3 |q_e|^{-c/12} = \prod_{e=1}^3 |q_e|^{-1/8}, \quad (8.21)$$

and the nine-superfield multiplier is

$$\prod_{e=1}^3 |q_e|^{-9/8}. \quad (8.22)$$

These optional constants must be applied consistently to both channels; they are not hidden inside A , P_X , or $\mathcal{F}_\psi$. If the Human-Note definition of L_e already contains the Casimir shift, (8.21)–(8.22) must not be inserted a second time.

At the `o0243-periodmatched` point, the numerical multiplication of the three normalized factors is

channel	$\mathcal{G}_X$	$ P_X ^2$	$ \mathcal{F}_\psi ^2$	$Z_{X+\psi}$	$Z_{X+\psi}^9$
theta	0.215006349267368	1.00000166309736	1.00258782998132	0.215563107646015	$1.00500893632052 \cdot 10^{-6}$
glasses	1.16308106016403	1.33846154450516	2.69778130216318	4.19974210080833	$4.06446695711553 \cdot 10^5$

These are raw plumbing-frame values with the normalization ledger above. Their inequality is not a failure of the resummation: the channel-change and Weyl factors have deliberately not been applied.

9 Recomputed Q_L with the plumbing denominator

The existing Human-Note-sign super-Liouville numerators at `o0243-periodmatched` can be reused: changing the independent physical free-theory denominator does not change a conformal block, structure constant, or momentum-quadrature node. Replacing the old period-matrix shortcut by (8.11) and (8.18), at the common cutoff $M = 24$, gives

channel	direct $Z_{X+\psi}^{\text{pl}}$	old shortcut	direct/old	Q_L multiplier
theta	0.215563107646015	0.431128572561718	0.499997266163927	512.025195722
glasses	4.19974210080833	8.39948420161306	0.500000000000214	511.999999998

The factor is fixed by the displayed charge convention. If $q = \exp(2\pi i\Omega)$, then for $h(\alpha) = \alpha^2/2$,

$$\int_{\mathbb{R}^g} d^g \alpha e^{-2\pi \alpha^T \text{Im } \Omega \alpha} = 2^{-g/2} \det(\text{Im } \Omega)^{-1/2}. \quad (9.1)$$

At $g = 2$, this is the factor $1/2$ that was absent from the old displayed shortcut. Raising one physical superfield to the ninth power therefore multiplies the absolute Q_L by approximately $2^9 = 512$. The small theta displacement from exactly 512 is the previously observed difference between the finite plumbing resummation and the shortcut, not a failure of the cutoff convergence.

The recomputed values are

R	N	$Q_L^\Theta/Q_L^{\text{Gl}}$	$Q_L^\Theta/Q_L^{\text{Gl}} - 1$
0	4	1.08934489273	0.08934489273
3	4	1.08996087620	0.08996087620
4	4	1.09091076293	0.09091076293
4	6	1.04478367268	0.04478367268

For the best local setting $(R, N) = (4, 6)$,

$$Q_L^\Theta = 7.23725849554 \times 10^{-12}, \quad Q_L^{\text{Gl}} = 6.92704019478 \times 10^{-12}. \quad (9.2)$$

A subsequent full local rerun of all four (R, N) settings with the direct denominator wired into the evaluator reproduced every numerator and every recombined Q_L exactly as stored in `Data Set/ns_genus2_theta_glasses_hatc9_human_note_plumbing_free_2026-08-25.json`.

Thus the new physical-Majorana and charged-boson plumbing computation changes the absolute normalization substantially but does not remove the channel discrepancy: the current ratio differs from one by 4.478%. The free denominator itself is stable well beyond 10^{-6} : from $M = 20$ to $M = 24$, the relative changes are below 3×10^{-15} in theta and 2×10^{-13} in glasses. The remaining visible movement is in the super-Liouville numerator, especially its momentum quadrature.

10 Audit and removal of the apparent four-percent discrepancy

The completed corrected-lift calculation at $(R, N) = (24, 10)$ first appeared to give

$$\frac{Q_L^\Theta}{Q_L^{\text{Gl}}} = 1.04056522832 \quad (10.1)$$

after replacing the old denominator by the direct plumbing denominator. This is not a free-denominator cutoff effect. The changes from $M = 20$ to $M = 24$ are below 2×10^{-13} , and replacing the direct denominator by the old period-matrix shortcut moves (10.1) by only 4.9×10^{-5} relatively. The available cutoff axes likewise show that the $R = 24$ recursion is stable and that the remaining $N = 10$ quadrature movement is at the few-times- 10^{-4} scale, not four percent.

The origin is instead a coefficient-convention mismatch at the interface between the real BRY super-Liouville data and the graded factorization in the Human Note. This is fixed by comparing definitions, not by fitting (10.1). For positive continuum momenta, the implementation `ns_tilde_structure_constant` returns BRY's real coefficient of the locally normalized top component

$$W_{\text{BRY}} = G_{-1/2} \tilde{G}_{-1/2} V, \quad \langle W_{\text{BRY}}, W_{\text{BRY}} \rangle = +D(2h)^2,$$

where D is the matched primary two-point normalization. The Human Note instead sews the ordered tensor state

$$w_{\text{HN}} = (G_{-1/2} V) \otimes (\tilde{G}_{-1/2} \tilde{V}).$$

It declares left- and right-moving odd operators to anticommute and uses the graded tensor-product BPZ pairing. Therefore its definition gives directly

$$\langle w_{\text{HN}}, w_{\text{HN}} \rangle_{\text{HN}} = -D(2h)^2.$$

After matching the primary normalization and the chiral three-form normalization, the boundary dictionary is consequently

$$C_{\text{HN}}^{(0)} = C_{\text{BRY}}, \quad C_{\text{HN}}^{(1)} = \sigma i \tilde{C}_{\text{BRY}}, \quad \sigma \in \{+1, -1\}. \quad (10.2)$$

Thus the BPZ comparison fixes the squared map, but not its square-root branch. The implementation chooses $\sigma = +1$; $\sigma = -1$ gives the same genus-two result because every term contains two pants coefficients. The odd coefficient product supplies

$$(C_{\text{HN}}^{(1)})^2 = -\tilde{C}_{\text{BRY}}^2$$

in theta, and the same minus sign in $C_{\text{LBL}}^{(1)} C_{\text{RRB}}^{(1)}$ in glasses. This sign is *not* a change to the Human Note. Its explicit descendant/Koszul sign $(-1)^a$ remains exactly as written; (10.2) only converts the imported BRY coefficient into that convention before the Human-Note formula is applied.

This scope distinction is essential. BRY-native Type 0B sphere correlators, including the G, H, J formulas and PCO amplitudes, retain the real $\tilde{C}_{\text{BRY}}$ with no additional i . Equation (10.2) is applied exactly once only when BRY data enter a Human-Note graded sewing sum. It is also unrelated to both the physical free Majorana in the denominator of Q_L and the auxiliary Majorana in the double-*Virasoro* construction. The removal of the four-percent discrepancy corroborates this dictionary; it

is not the argument that determines it. Fixing σ in an amplitude with an odd number of odd pants would require one explicitly ordered component matrix element.

The (24, 10) shard decomposition makes the diagnosis numerical and unambiguous:

channel	even contribution	signed odd contribution before (10.2)
theta	$7.27241679849612 \times 10^{-18}$	$-2.77708574019467 \times 10^{-20}$
glasses	$2.88444939646816 \times 10^{-6}$	$-6.87808460834071 \times 10^{-8}$

Inserting (10.2) reverses the displayed odd contributions while retaining the Human-Note sewing sign. With the direct plumbing denominator this gives

$$\begin{aligned} Z_L^\Theta &= 7.30018765589807 \times 10^{-18}, & Q_L^\Theta &= 7.26380372559184 \times 10^{-12}, \\ Z_L^{\text{Gl}} &= 2.95323024255156 \times 10^{-6}, & Q_L^{\text{Gl}} &= 7.26597183274289 \times 10^{-12}, \end{aligned}$$

and therefore

$$\boxed{\frac{Q_L^\Theta}{Q_L^{\text{Gl}}} = 0.999701608099, \quad \frac{Q_L^\Theta}{Q_L^{\text{Gl}}} - 1 = -2.98391901 \times 10^{-4}.} \quad (10.3)$$

Thus the convention repair removes 99.26% of the previous residual. It also improves monotonically in the inexpensive local quadrature check:

R	N	corrected $Q_L^\Theta/Q_L^{\text{Gl}} - 1$
4	4	$+4.84740320 \times 10^{-2}$
4	6	$+3.83177997 \times 10^{-3}$
4	8	$+1.40461919 \times 10^{-3}$
24	10	$-2.98391901 \times 10^{-4}$

The remaining 2.98×10^{-4} is consistent with the unresolved momentum-quadrature tail at $N = 10$; it is no longer evidence for a four-percent plumbing, free-theory, or recursion failure. A fresh common $R = 24$, $N = 12$ or higher run is still required before claiming 10^{-6} stabilization.

11 Executable implementation and checks

The theta and glasses fermion determinants, charged-boson Gaussians, and complete one- and nine-superfield expressions are implemented in

`Code/genus_2/physical_free_plumbing_resummation.py`

The focused test compares (4.5) and (7.4) with direct physical-Majorana Fock/Pfaffian sums for three lift assignments. It also compares (5.6) and (7.5) with the independent primitive-word boson products:

`Code/genus_2/test_physical_free_plumbing_resummation.py`

All eleven focused checks pass. The implementation reports separately the fermion determinant, Heisenberg determinant, charged-boson Schur complement, two-by-two loop matrix, one complete superfield, and its ninth power.

The fast production entry point

physical_superfield_plumbing_partition

uses the bosonic Gaussian vacuum determinant directly. Primitive-word enumeration is retained only as an independent test oracle.

At the o0243-periodmatched plumbing point, using only the listed plumbing coordinates and Human-Note lifts, consecutive-cutoff stabilization below 10^{-6} occurs at

channel	common determinant cutoff	local time
theta	$M = 8$	about 2 ms
glasses	$M = 10$	about 10 ms

The converged raw ingredients in the $h = \alpha^2/2$ charge convention are

Theta.

$$\begin{aligned}
\mathcal{F}_\psi &= 1.00129307876105 - 2.01628153784 \cdot 10^{-5}i, \\
P_X &= 1.00000083154833 - 3.00494124702 \cdot 10^{-8}i, \\
\mathcal{G}_X &= 0.215006349267368, \\
Z_{X+\psi} &= 0.215563107646015.
\end{aligned}$$

Glasses.

$$\begin{aligned}
\mathcal{F}_\psi &= 1.64249239061865 + 2.21185456733 \cdot 10^{-4}i, \\
P_X &= 1.15691889842888 - 4.54930513585 \cdot 10^{-4}i, \\
\mathcal{G}_X &= 1.16308106016403, \\
Z_{X+\psi} &= 4.19974210080833.
\end{aligned}$$

The timings are single-process local timings after import and are meant only to show the computational order. The production determinant computation scales as $O(M^3)$; primitive-word enumeration is no longer part of the production denominator.